

A Mass-Limited Rate Barrier for Interior Annular Norms

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Abstract

We turn the odd-anchor mass demand into an unconditional norm lower bound for every modulus-capped complement. Given squared spectral mass M , choose the first odd order at which the cap Mq^{n-2} falls below half of the exact deterministic anchor. That coefficient alone forces both the $H^\infty(\rho)$ and $H^2(\rho)$ mismatch norms to be at least $C_\rho M^{-\kappa(\rho)} / \log M$, where $\kappa(\rho) = \log(1/(q_*\rho)) / \log(q_*/q)$. Under $M_\sigma \leq \sigma^{-1}$, no interior annular norm can decay with a power of σ strictly larger than $\kappa(\rho)$. The lower bound tends to zero and therefore does not exclude actual annular convergence.

1 Mass-dependent odd witness

Fix $R < \rho < \rho_* = q_*^{-1}$ and let

$$g(z) = \sum_{n \geq 2} \frac{\tau_n - a_n}{n} z^n, \quad \tau_n = \sum_j \mu_j^n, \quad |\mu_j| \leq q, \quad M = \sum_j |\mu_j|^2.$$

For odd n , $a_n = q_*^n / (1 + \lambda^{-n})$. Let $N(M)$ be the least odd $n \geq 3$ such that

$$Mq^{n-2} \leq \frac{1}{2}a_n.$$

The logarithmic implementation uses this exact inequality and does not replace it by an asymptotic proxy.

2 Strict and asymptotic lower bounds

Theorem 2.1. *For $X = H^\infty(\rho)$ or $H^2(\rho)$,*

$$\|g\|_X \geq \frac{(q_*\rho)^{N(M)}}{2N(M)(1 + \lambda^{-N(M)})}.$$

Moreover, there are constants $C_\rho > 0$ and $M_{0,\rho}$ such that for $M \geq M_{0,\rho}$,

$$\|g\|_X \geq C_\rho \frac{M^{-\kappa(\rho)}}{\log(e + M)}, \quad \kappa(\rho) = \frac{\log(1/(q_*\rho))}{\log(q_*/q)}.$$

Proof. At $N = N(M)$, the spectral cap gives $|\tau_N| \leq Mq^{N-2} \leq a_N/2$, hence $|\tau_N - a_N| \geq a_N/2$. A single coefficient of either annular norm gives the first display.

Put $r = q_*/q > 1$ and $y = q_*\rho < 1$. The defining inequality for N and its failure two orders earlier imply

$$N(M) = \frac{\log M}{\log r} + O(1).$$

Therefore $y^{N(M)} = \Theta_\rho(M^{\log y / \log r}) = \Theta_\rho(M^{-\kappa(\rho)})$, while $N(M) = \Theta(\log M)$. Absorb the bounded factors into C_ρ . $\square$

Corollary 2.2 (small-noise power ceiling). *For the actual modulus complement, $M_\sigma \leq \sigma^{-1}$. Hence, after adjusting a constant for bounded masses,*

$$\|g_\sigma\|_X \geq C_\rho \frac{\sigma^{\kappa(\rho)}}{\log(e + 1/\sigma)}.$$

In particular $\|g_\sigma\|_X = O(\sigma^\beta)$ is impossible for every $\beta > \kappa(\rho)$.

Proof. The function $M^{-\kappa}/\log(e + M)$ is eventually decreasing. Apply Theorem 2.1 with $M_\sigma \leq \sigma^{-1}$. If M_σ stays bounded along a subsequence, the exact witness bound is bounded below by a positive constant and the conclusion is stronger. $\square$

3 Numerical protocol and boundary

At $\rho = 1.41$,

$$\kappa(1.41) = 0.035045705260961 \dots$$

The computation uses the exact least odd witness at $M = 10^{12}$ for $\rho = 1.405, 1.41, 1.42$ and records the certified coefficient lower bounds. It does not treat the asymptotic expression as a unit-constant global bound.

Remark 3.1. The theorem rules out overly fast convergence, not convergence itself. It uses only the actual modulus cap, squared spectral mass, and exact odd anchors; it does not identify the head or close any determinant gate. Gates A–E remain false/open.